

Who this helps

Twelve situations, each with the data that makes it real

The test for every one of these: it has to be something a person in Nagpur is *actually* wrong about right now, where being wrong costs them something, and where the correct answer already exists in data nobody has connected to them.

1. The ₹18,000 water purifier that solves the wrong problem

A family buys an RO because a salesman held up a TDS meter. But RO removes *dissolved solids*. If you're on treated municipal supply, dissolved solids were rarely your problem — the risk in an intermittent-supply zone is bacteriological, entering through the distribution network, and re-entering downstream of your purifier in the storage tank anyway.

Meanwhile a family on a borewell in a block where CGWB found nitrate above the limit **has** a dissolved-solids problem, and nitrate is exactly what RO removes — and nobody told them.

The same appliance is a waste in one part of Nagpur and genuinely necessary in another. The deciding factor is which source you're on, and that's the one thing nobody tells you.

2. "Go deeper, it'll be cleaner" — an assumption nobody has checked

Everyone believes a deeper borewell is safer water. In CGWB's Nagpur district sampling, the deep aquifer exceeded limits *more* often than the shallow one:

	Nitrate above limit	Fluoride above limit
Shallow aquifer	35 of 263 — 13%	6 of 263 — 2.3%
Deep aquifer	17 of 84 — 20%	4 of 84 — 4.8%

State this carefully — it is not a proven finding. The samples are unpaired, the sample sizes are very different (263 vs 84), and nothing establishes they come from the same villages. A two-proportion test gives $z \approx 1.55$, $p \approx 0.12$ — not significant, and the confidence intervals overlap heavily.

The honest version, which is still worth saying: *"In this dataset, deep samples exceeded the nitrate limit more often than shallow ones. The samples aren't paired, so this is a question worth asking — not a conclusion."*

A family about to spend ₹1.5 lakh drilling deeper for safety is acting on an assumption that nobody has actually tested for this district. That alone is worth knowing before you spend.

3. Nagpur's oranges are being killed slowly by their own irrigation water

Citrus is among the more salt-sensitive crops there is. CGWB found electrical conductivity above **2250 $\mu\text{S}/\text{cm}$ — the irrigation limit — in 5.3% of shallow samples**, and at Belgaon in Umred it measured **4920 $\mu\text{S}/\text{cm}$** , over double the limit.

An orange grower on that water loses yield every single year and attributes it to disease, weather, or bad luck. Salinity damage is gradual and looks like everything else.

The city's signature crop, damaged by its own water, and nobody connects the two because nobody puts the number in front of the farmer in a form that means anything.

4. Which side of the outfall are you on?

There are **9 nalla outfalls into the Nag and 3 into the Pili**. Two farms 400 metres apart, one just above an outfall and one just below, are drawing completely different water. Measured BOD goes from **4-13 mg/L at the Ambazari origin to an average of 57 at Bhandewadi Bridge**.

Your position relative to the nearest outfall is the single biggest thing determining your water quality — and it is completely invisible to you.

5. 99.6% is an excellent result and a useless answer

OCW reports 99.6% of samples fit in 2025–26. That's genuinely good, and the utility deserves credit for it.

But 0.4% of a year's sampling across 68 reservoirs still means failures happened — each in a specific place, on a specific day. Nobody living near those places was told, because the number is published city-wide.

This isn't a story about negligence. It's a story about a good number published at the wrong resolution to be useful to anyone.

(That distinction matters — a municipal jury will shut down instantly if you sound like you're accusing the utility.)

6. The last twenty metres nobody owns

The utility's responsibility ends at your meter. Your knowledge ends at your tap. In between sit the building's underground sump and the overhead tank — cleaned rarely, if ever, and mandatory precisely *because* supply is intermittent. Perfectly good water can arrive and fail in the last twenty metres, and there's no one whose job it is to know.

7. Your water has a date on it, and nobody re-tests

A borewell tested in September is not the same borewell in May. As levels fall, concentrations rise. **65 of 72 CGWB monitoring stations in the district show declining pre-monsoon water levels**, some by metres per year.

A test result from three years ago is treated as a permanent fact about a well. It isn't.

8. The underpass problem, which is specific to two-wheelers

Nagpur's 2023 flood came from 109 mm in a night. The characteristic killer isn't an open road — it's an **underpass**: a low point that fills while everything around it stays dry, so it looks passable right up until you're in it, and the water is opaque so you cannot judge depth from the edge. Maps reroutes you once cars have already stopped. That is too late for someone on a two-wheeler who left home at 7 AM.

The warning has to arrive the night before, and it has to be about your route specifically.

9. Getting the stock off the floor, and proving it later

A ground-floor shop in a chronic clog spot has two separate problems. Three hours' warning gets the stock onto shelves. And after 2023, **₹5,00,000 was announced for affected businesses** — but proving you were affected requires a record that doesn't exist.

One system solves both: the warning before, and the timestamped record after.

10. The official who wants to be able to say no

A ward officer has one desilting budget and thirty streets asking for it. The current selection method is pressure and memory — not because anyone is corrupt, but because **no ranked list exists**.

A defensible ranking isn't a threat to that officer, it's cover: "I can show you exactly why your street is twelfth this year." That's why they'd actually use it.

11. ₹266.63 crore was spent. Did it work?

That's the mitigation package announced after the 2023 flood. Money was spent, works were done. Whether any specific spot is actually better now is unanswerable, because **nobody recorded the before**.

Asking "was it fixed?" thirty days later doesn't just check a repair. It creates the baseline that doesn't currently exist.

12. Not everyone can afford "probably fine"

For a household with an infant, an elderly relative, or someone immunocompromised or on dialysis, "the water is generally okay" is not an adequate answer — but they currently have no way to be more careful than anyone else, because everyone gets the same single city-wide number.

Risk isn't distributed evenly, and neither should the information be.

The pattern underneath all twelve

Every one is a person making a real decision about water — where to live, what to plant, whether to drill, which road to take, where to spend a budget — **with no information, when the information already exists.** It's aggregated into a city-wide percentage, buried in a government PDF, or sitting in someone's memory of last year's monsoon.

We are not measuring new things. We are connecting what is already measured to the person it affects, at the resolution they actually live at.